

China Semiconductors

Global Memory Interface Chip: Competitive dynamics deep dive



Qingyuan Lin, Ph.D.
+852 2123 2654
qingyuan.lin@bernsteinsg.com



Francis Ma
+852 2123 2626
francis.ma@bernsteinsg.com



Kai Zhang
+852 2123 2665
kai.zhang@bernsteinsg.com

We highlighted in our [memory interface chip industry deep dive](#) that all three major sector vendors (Montage, Renesas, Rambus) will benefit from the agentic AI boom, in the report we provide a deep dive to the competitive dynamics with a focus on Rambus (RMBS, not covered). Our industry model can be downloaded [here](#).

Rambus derives revenue from memory interface chips, DRAM-related IP royalties, and computing-chip IP licensing, exposed to three AI-driven growth engines that should support a strong revenue growth through 2030. Each segment is leveraged to a major secular AI theme: memory interface chips benefit from the server CPU Renaissance driven by agentic AI; DRAM royalties are tied to memory tech migration; and computing-chip IP licensing benefits from growing share of AI ASICs. Together, these businesses provide broad exposure to the core infrastructure layers underpinning AI development. We project the memory interface chip market alone to reach nearly USD 20Bn by 2030. ASIC will account for over [two-thirds](#) of CoWoS-based XPU shipments in 2027. Both highlight the magnitude of the opportunities available to Rambus.

However, each of Rambus's core businesses faces competitive constraints that are likely to keep growth below Montage. In memory interface chips, we believe its historical litigation-driven relationship with memory manufacturers limits its ability to gain shares from Montage and Renesas. In DRAM royalties, long-duration licensing contracts provide stable cash flow but inherently cap further growth. In IP for computing chips, Rambus competes against both vertically integrated AI ASIC designers with extensive in-house IP portfolios and ecosystem leaders such as Cadence and Synopsys, whose IP offerings are tightly bundled with EDA tools and foundry workflows. Consequently, Rambus is largely positioned to serve AI chip startups rather than first-tier hyperscaler ASIC programs.

A broader technology portfolio combined with slower revenue scaling should translate into a slower operating leverage than Montage. Rambus invests across memory interface chips, memory controllers, interconnect IP, security technologies, and applications in diverse end markets, resulting in annual R&D spending roughly 40-50% higher than Montage, whose tech portfolio is more focused. While this broad portfolio expands Rambus's addressable market, it also requires a significantly larger engineering organization. Because revenue growth is moderated by the fierce competition discussed above, Rambus generates less revenue per dollar of R&D spending than Montage. Consequently, we expect operating leverage to emerge more gradually for Rambus, leading to slower net income growth than Montage.

We expect Rambus to grow slower than Montage in both revenue and earnings as a result. Market has been trading Rambus at a discount vs. Montage, given slower expected revenue and earnings growth. Nevertheless, a business that rides on 65% TAM growth remains an attractive growth asset.

BERNSTEIN TICKER TABLE

Ticker	Rating	Cur	23 Jul 2026	Price Target	TTM	Reported EPS				Reported P/E (x)		
			Closing Price		Rel. Perf.	Cur	2025A	2026E	2027E	2025A	2026E	2027E
6809.HK (Montage)	O	HKD	311.80	520.00	NA	CNY	1.97	3.21	5.68	136.7	84.0	47.4
688008.CH (Montage)	O	CNY	225.50	400.00	142.3%	CNY	1.97	3.21	5.68	114.5	70.3	39.7
ASIAx			1,880.30									

O - Outperform, M - Market-Perform, U - Underperform, NR - Not Rated, CS - Coverage Suspended

Source: Bloomberg, Bernstein estimates and analysis.

INVESTMENT IMPLICATIONS

We rate **Montage Outperform**, with A-Share TP at CNY 400 and H-Share TP at HKD 520.

Table Of Contents

Company overview.....	3
Memory interface chips: accelerating TAM growth.....	5
Royalties of IP inside the DRAM module: stable but low-growth.....	10
Licensing IP inside the computing chips: attractive end market, but intense competition.....	11
Moderate operating leverage and earnings growth restricted by R&D intensity.....	14

DETAILS

COMPANY OVERVIEW

COMPANY INTRODUCTION

Rambus is a Silicon Valley veteran that has reinvented itself twice - from a memory technology inventor into a patent-licensing house, and over the past decade into a fabless as memory interface chip supplier. Founded in 1990 by Stanford engineers Mike Farmwald and Mark Horowitz and listed on Nasdaq in 1997, the company first rose to prominence with RDRAM, a proprietary high-speed memory interface adopted by Intel around the Pentium 4 era. When the industry standardized on DDR SDRAM instead, Rambus spent the following decade monetizing its patent estate through litigation and licensing against the major DRAM manufacturers, a history we detail in the Royalties business section below. The pivot back to silicon began in 2016 with the acquisition of Inphi's memory interconnect business, which brought DDR4 buffer chips, and accelerated under current CEO Luc Seraphin, who was appointed in 2018 and refocused on memory interface chips.

Today, Rambus operates three revenue pillars: memory interface chips, where it ranks third globally with ~21% share of the core market in 2024; patent royalties from DRAM manufacturers; and silicon IP licensed to computing chip designers. The company generated USD 708Mn of revenue in 2025, up 27% YoY.

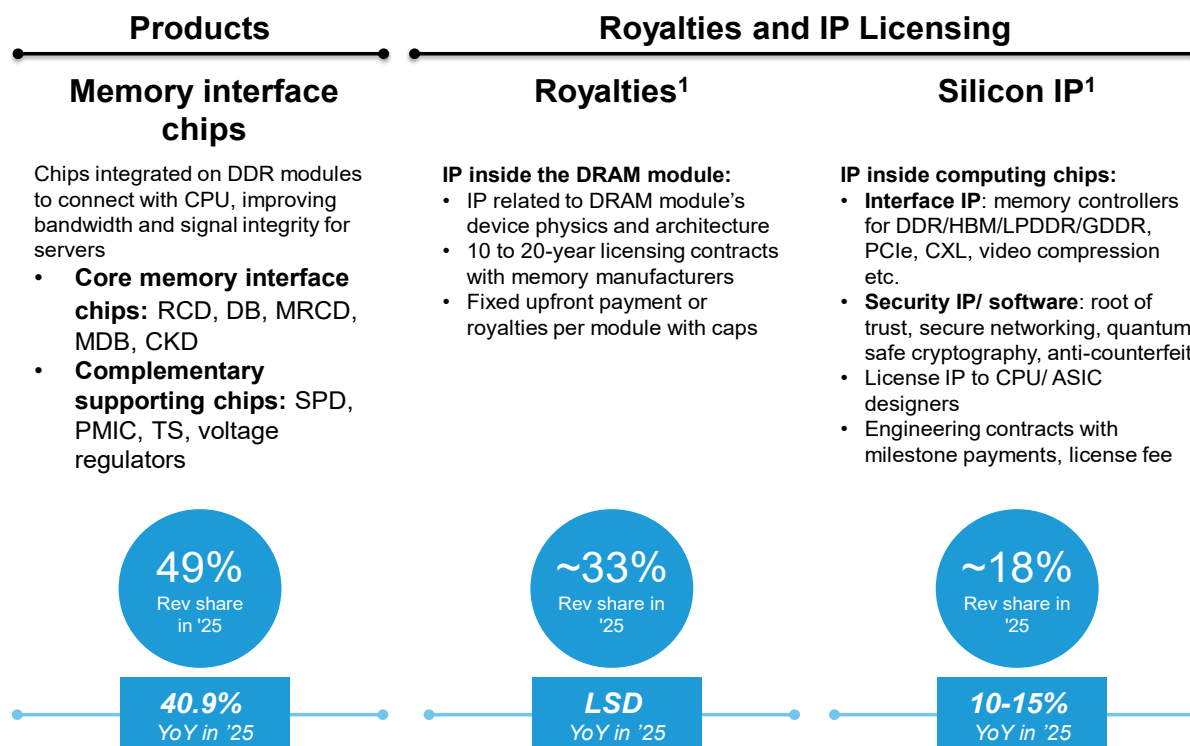
PRODUCT OFFERINGS

Rambus runs two broad business segments, Products and Royalties/ IP Licensing, which are reported in three lines in its financial statements: Product revenue, Royalty revenue, and Contract & other revenue.

The Product line consists of memory interface chips integrated in DDR modules to connect with server CPU, improving bandwidth and signal integrity for servers. This spans the core memory interface chips (RCD, DB, MRCD, MDB, and CKD) and the complementary supporting chips (SPD Hub, PMIC, temperature sensors, and voltage regulators). Product revenue contributed 49% of total revenue in 2025 and grew 40.9% YoY, making it the company's largest and fastest-growing line.

The Royalties/ IP Licensing segment monetizes two distinct groups of IP for two distinct customer bases, memory manufacturers and computing chip designers. IP inside the DRAM module contributed ~33% of 2025 revenue and grew in LSD YoY. These patents are mainly related to the module's device physics and architecture, licensed to memory manufacturers under 10-20 year contracts structured as fixed upfront payments or per-module royalties with caps. Silicon IP contributed the remaining ~18% of 2025 revenue and is guided to grow at 10-15% YoY. Silicon IP licenses IP inside computing chips to ASIC/ SoC designers, including memory controllers IP for DDR/ HBM/ LPDDR/ GDDR and interconnect IP for PCIe/ CXL, as well as security IP spanning root of trust, secure networking, quantum-safe cryptography, and anti-counterfeit. These engagements combine license fees with engineering contracts carrying milestone payments.

EXHIBIT 1: Rambus has two primary business segments but reports financial statements in three lines. Two business segments ride on the wave of CPU Renaissance and AI ASIC respectively



The revenue mix and growth rate numbers are operational approximations per management commentary, not formal reporting segments
Source: company reports, Bernstein analysis

BUSINESS MODEL

The two segments run on fundamentally different business models - one sells chips, the other licenses IP rights.

The memory interface chip Product business is a classic fabless model: Rambus designs the chipsets, outsources fabrication to foundry partners, and sells physical chips to Samsung, SK hynix, and Micron. Revenue scales directly with module volumes and content per chipset, giving the business full exposure to the server memory upgrade cycle.

The Royalties and Silicon IP businesses are licensing models: Rambus monetizes intangible assets through long-duration patent contracts and design-win-based IP licenses with engineering milestones. Licensing revenue is extremely high-margin and recurring, but its growth is bounded by contract structures.

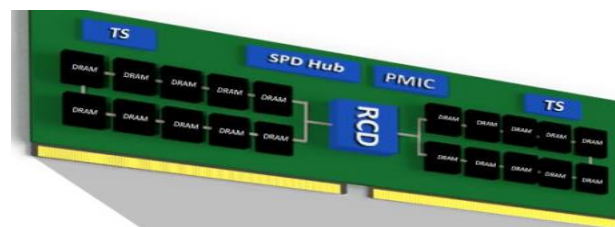
Overall, the Product business carries the growth story, while licensing provides the stable cash flow that funds R&D.

MEMORY INTERFACE CHIPS: ACCELERATING TAM GROWTH

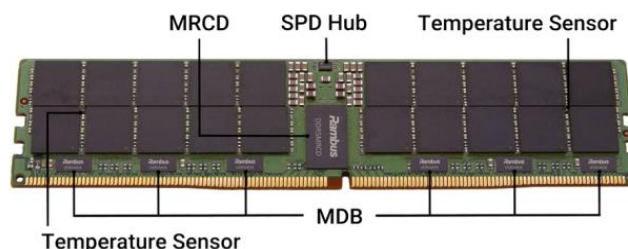
As we have elaborated in [our memory interface chip primer](#), these chips are a core beneficiary of the Agentic AI era, and **Rambus enjoys the full strength of this industry beta**. Agentic AI is structurally reshaping demand for server CPUs, DDR memory, and the interface chips connecting them. As workloads shift from GPU-centric training toward inference-heavy agentic deployments, the CPU reclaims its role as orchestrator of multistep, sequential tasks.

EXHIBIT 2: Providing memory interface chips for RDIMM & MRDIMM alongside Montage, Rambus is the only supplier of SOCAMM2 interface chips today

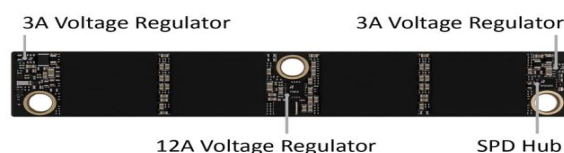
Interface chips
for RDIMM



Interface chips
for MRDIMM



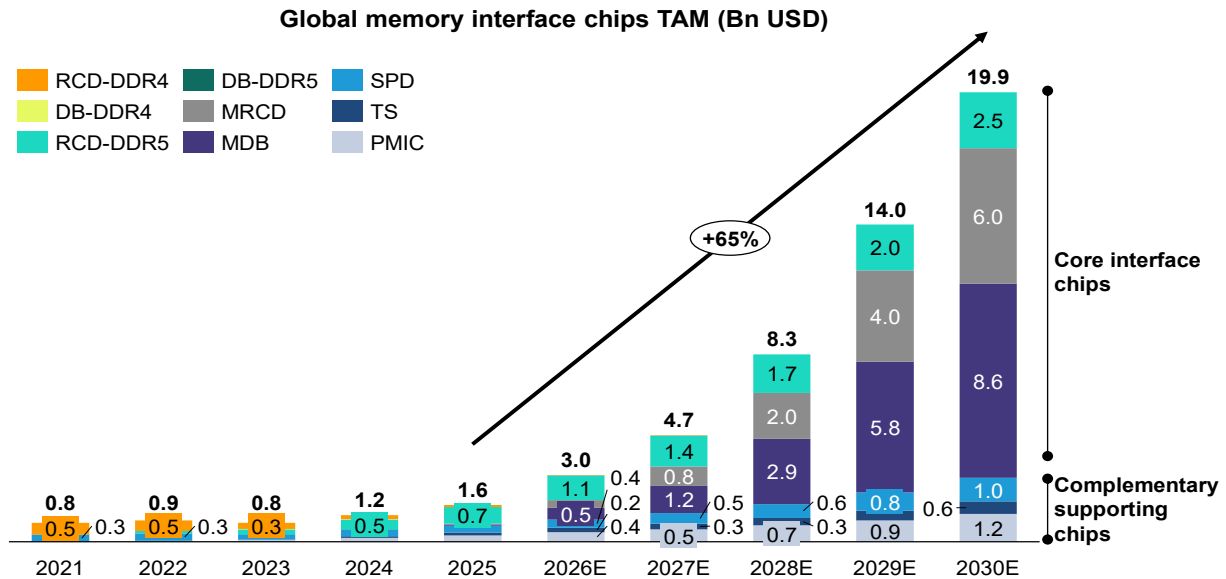
Interface chips
for SOCAMM2



Source: company reports, Bernstein analysis

We project the global memory interface chip TAM to grow at ~65% CAGR'25-'30 to reach USD 20Bn in 2030. The strong momentum will be driven by three compounding tailwinds: rising server CPU volumes, more DRAM package per CPU, and higher chipset value per module through the MRDIMM upgrade. MRDIMM is the single largest contributor: we project MRCD and MDB to account for 73% of 2030 TAM, as interface silicon content per MRDIMM module runs roughly 10x that of RDIMM. Importantly, the downside is well protected: even in a bear-case scenario on CPU shipments and MRDIMM penetration, the 2030 TAM still reaches our prior-version projection in \$ 8Bn.

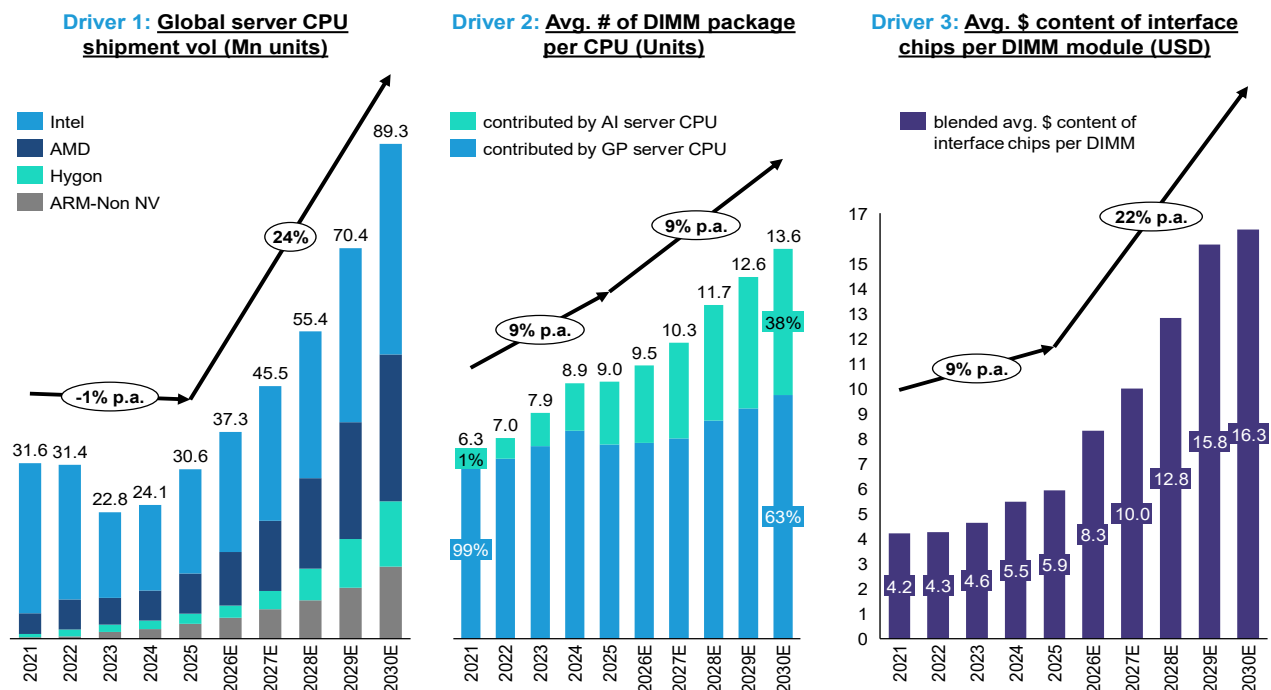
EXHIBIT 3: We raise our global memory interface TAM projection to USD 20Bn in 2030, with MRDIMM's MRCD and MDB contributing 73%, driven by significantly higher chipset value per module



Our TAM forecast does not include memory interface chips used in NVIDIA proprietary CPUs, such as voltage regulators and PMICs. We expect the impact to be only LSD, as NVIDIA CPUs account for a small portion of server CPU shipments and the related components are of relatively low value.

Source: Mercury, TrendForce, Gartner, Bernstein analysis and estimates

EXHIBIT 4: The expansion of the TAM is driven by three core factors, each of which is expected to accelerate in the second half of this decade relative to the past 5 years



Source: Gartner, Mercury, Bernstein analysis and estimates

EXHIBIT 5: 2030 TAM Sensitivity Analysis: Even in a bear-case scenario, the global memory interface chip TAM reaches our prior-version projection, reinforcing a well-protected downside for all memory interface chip suppliers

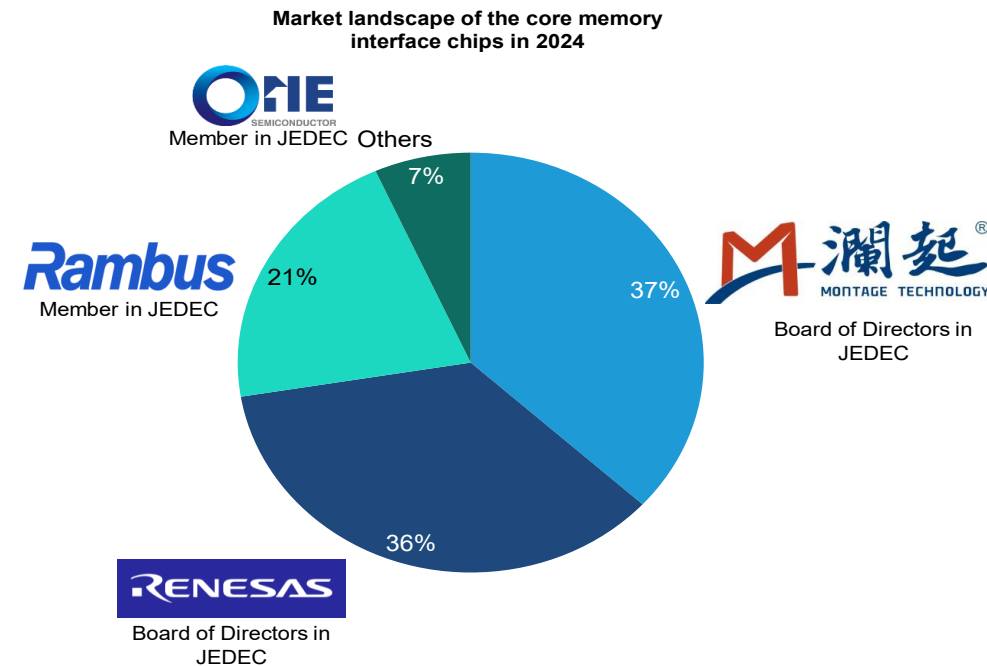
2030 TAM sensitivity analysis (Bn USD)		Server CPU shipment in 2030 (excl. NV, Mn units)						
		60	70	80	89	100	110	120
MRDIMM penetration in 2030	10%	8	9	10	12	13	14	16
	15%	10	11	13	14	16	18	19
	20%	12	13	15	17	19	21	23
	25%	13	16	18	20	22	24	27
	30%	15	18	20	23	25	28	30
	35%	17	20	23	25	28	31	34
	40%	19	22	25	28	32	35	38

Source: Bernstein analysis and estimates

Within this expanding market, Rambus competes from the third position in a textbook oligopoly and holds a differentiated footprint across module form factors. The top three players consolidated 90% of the core memory interface chip market in 2024, with Rambus as ~21% behind Montage and Renesas. For DDR5 RDIMM RCD, Rambus management claims that the company is expanding its market share continuously. The company has also closed its historical gap in complementary supporting chips through sequential in-house development since 2022 to pursue a “full chipset” strategy, as management noted that customers increasingly want the whole chipset from one supplier for interoperability reasons at high speed. Now Rambus offers a fully in-house chipset for both RDIMM and MRDIMM, and the management has guided the complementary supporting chips to contribute mid-double-digit percentage of revenue by year end 2026. Notably, Rambus is today the only supplier of SOCAMM2 interface chips, giving it a first-mover position in emerging socket.

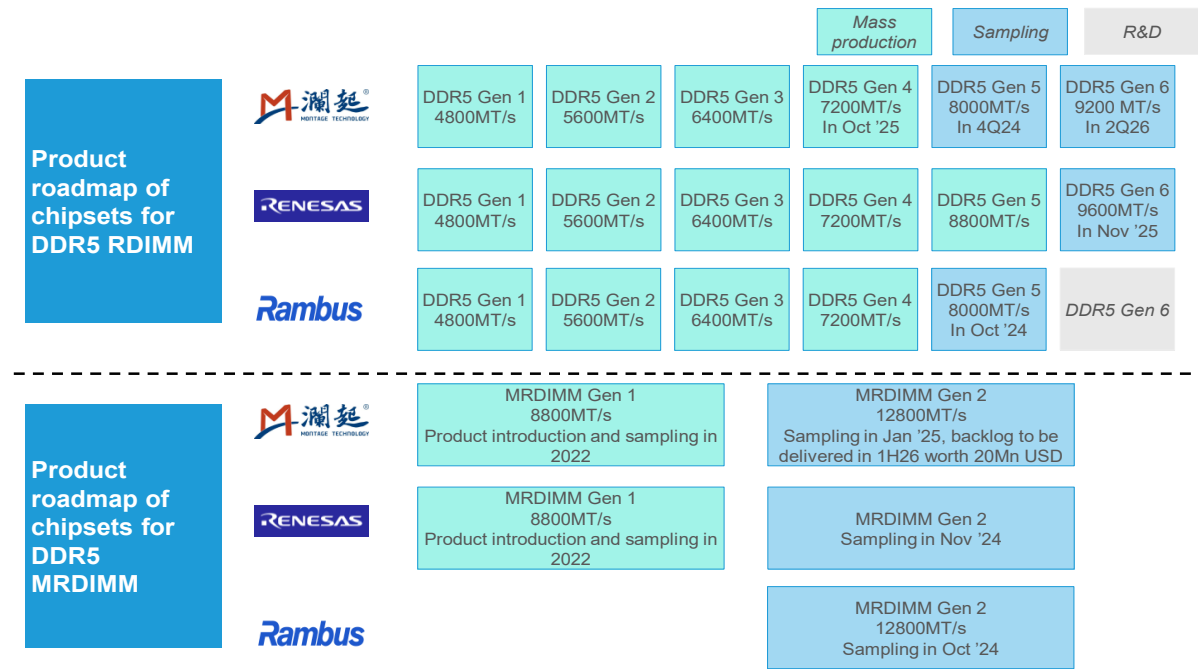
As a late entrant to MRDIMM, Rambus has material room to expand share in the segment where the TAM growth is concentrated. Skipping the MRDIMM Gen 1, Rambus directly resources to Gen 2. Starting from a currently almost negligible share base, every Gen 2 design win is incremental: we model Rambus’s MRDIMM interface chipset share rising from 2% in 2026 to 18% by 2030, conservatively assuming its long-term market share in MRDIMM chipset lower than that in RDIMM chipset. Meanwhile, management has also guided for continued share gains in complementary supporting chips, where its portfolio build-out is now complete. In short, the company’s share trajectory in the highest-growth product category points upward from a low base, a favorable setup relative to competitors defending high shares.

EXHIBIT 6: **The core memory interface chips market is a textbook oligopoly, with top 3 players consolidating 90+% of market share. Rambus is ranked in the 3rd place.**



Source: company reports, Bernstein analysis and estimates

EXHIBIT 7: **Three industry leaders remain closely matched in DDR5 RDIMM and MRDIMM offerings. However, Rambus is the late entrant in MRDIMM memory interface chips**



Source: companies official websites, Bernstein analysis

However, we believe Rambus's market share is structurally capped, owing to its historical delicate relationship with memory manufacturers. Rambus's history is deeply rooted in patent enforcement and litigation against the DRAM makers,

who are the customers and co-validation partners that memory interface chip vendors depend on. While this strategy built a valuable intellectual property portfolio, we believe the legacy relationship may constrain Rambus's ability to achieve a dominant position. DRAM makers control module qualification and favor suppliers with whom they have deep co-development trust, the dual-sourcing preferences naturally distribute share toward the two competitors. We expect Rambus to continue gaining share as MRDIMM adoption accelerates, but we believe its long-term share in MRDIMM interface chips is likely to be restricted to the third position, given the competitive landscape and customer dynamics.

ROYALTIES OF IP INSIDE THE DRAM MODULE: STABLE BUT LOW-GROWTH

Two decades of patent litigation converted Rambus's IP estate into a recurring royalty annuity, the origin of today's second-largest revenue line. The campaign began with the Infineon suit in Aug 2000, the first major test of Rambus's DDR patent claims, and ran through the FTC's antitrust action over Rambus's conduct at the JEDEC standards body, removing the principal legal overhang and preserving the enforceability of the patent estate. A jury verdict in Rambus's favor against SK hynix and Micron in 2008 then set the stage for negotiated settlements with all three major DRAM makers: Samsung in 2010, and SK hynix & Micron in 2013. These settlements have since been renewed and extended into the 10-20 year licensing contracts that underpin the segment today.

EXHIBIT 8: Two decades of patent litigation converted Rambus's IP estate into recurring royalties revenue. But we think this delicate relationship with memory vendors may structurally restrict its share in interface chip sector

Counterparty / Case	Period	Forum	Core issue	Outcome	Legacy for Rambus today
Infineon Technologies	Aug 2000 - 2005	U.S. federal courts (E.D. Va.; Fed. Cir.)	Alleged infringement of Rambus patents by DDR SDRAM; Infineon counterclaimed fraud over JEDEC conduct	Settled in 2005 with a patent license agreement (terms per press reports)	First major litigation converted to license; template for later settlements
U.S. FTC, Docket 9302	2002 - 2008	Federal Trade Commission; D.C. Circuit	Antitrust: alleged deception of the JEDEC standards body while patenting standard-essential technology	FTC ruled against Rambus (2006); decision overturned on appeal (2008)	Removed the antitrust overhang; preserved enforceability of the patent estate
Hynix / Micron (coordinated trials)	2000s - Mar 2008	U.S. federal courts (N.D. Cal. and others)	Patent infringement and JEDEC-conduct defenses tried across coordinated proceedings	Jury verdict rendered March 26, 2008 in Rambus's favor on key issues	Set the stage for negotiated settlements with all three major DRAM makers
Samsung Electronics	Settled Jan 2010	Settlement agreement	Resolution of all outstanding patent claims	Agreement valued at ~\$900Mn, including a \$200Mn upfront payment and a \$200Mn purchase of Rambus stock (per contemporaneous press reports)	Largest single settlement; anchor licensee relationship, since renewed
Rambus v. Nvidia	2008 - 2010	U.S. ITC and federal courts	Alleged infringement of Rambus memory controller patents	Nvidia entered a patent license agreement with Rambus (2010, per press reports)	Extended licensing beyond DRAM makers to logic/GPU designers
SK hynix	Settled Jun 2013	Settlement / license renewal	Resolution of the long-running Hynix patent dispute	\$240Mn five-year license, structured as \$12Mn per quarter	Recurring royalty stream, since extended into a long-term agreement
Micron Technology	Settled Dec 2013	Settlement agreement	Resolution of the final major DRAM patent dispute (RDRAM/DDR era)	Seven-year agreement worth up to \$280Mn	Completed licensing coverage of all three major DRAM manufacturers

Source: SEC filings, Bernstein analysis

The result is a stable, annuity-like revenue stream, but one that management has guided to remain largely flat, offering limited growth. The agreements are structured as fixed upfront payments or per-module royalties with caps, which insulates the stream from DRAM cycle volatility but equally excludes Rambus from expanding in the current DRAM volume and pricing upcycle. We view this segment as a dependable cash generator that funds the company's R&D, but not a source of growth, a distinction that matters for the company-level growth profile discussed in the following sections.

LICENSING IP INSIDE THE COMPUTING CHIPS: ATTRACTIVE END MARKET, BUT INTENSE COMPETITION

The addressable market for memory controllers and interconnect IP used in computing chips is expanding rapidly, driven by the strong demand for custom AI ASICs. Our global semi team projects AI ASIC can reach 20% of the XPU market by value in 2027, up from 10% in 2026. Measured by shipment volume, Bernstein projects ASIC will account for over two-thirds of CoWoS-based XPU shipments in 2027. ASIC benefits from the industry's inflection from training to inference and will grow faster than the broader accelerator market. CSP, including Google, Microsoft, Amazon, and Meta, are accelerating ASIC adoption to improve performance per dollar, lower total cost of ownership, and reduce reliance on external GPU supply chains. Every one of these custom ASIC chips needs HBM and DDR controllers, interconnect, and security IP, which are exact portfolio Rambus licenses.

EXHIBIT 9: A diversified IP business spanning memory controllers, interconnect controllers, and security tech drives higher R&D intensity than Montage (1/2)

	Memory Controllers						Interconnect Controllers				Video Compression		
	DDR	LPDDR	GDDR	HBM			CXL		PCI Express		MIPI	VESA	FEC
Application Focus	4	4X/5X/5T	6/7	2/2E	3/3E	4/4E	2.0	3.1	6.3	7.0	CSI-2/ DSI-2	DSC/ VDC-M	DP/ HDMI
Data Center		R	R	R	R	R	R	R	R	R			
AI/Edge		R	R	R	R	R	R	R	R	R			
Automotive	R	R	R						R		R ★	R ★	R ★
Government & Defense	R								R				
IoT		R									R	R	R
Availability	Now	Now	Now	Now	Now	Now	Now	Now	Now	Now	Now	Now	Now

Source: company reports

EXHIBIT 10: A diversified IP business spanning memory controllers, interconnect controllers, and security tech drives higher R&D intensity than Montage (2/2)

	Root of Trust		Network Security			Cipher Engines	Crypto Cores		Provisioning	
Application Focus	Programmable	Firmware Controlled	MACsec	UET-TSS	IPsec TLS	IME	Crypto	TRNG	Infrastructure	Key Management
Data Center	  	N/A					 	 		
AI/Edge	   	  					 	 		
Automotive	 	 		N/A	N/A					
Government & Defense	  			N/A			  	 		
IoT	N/A	  	N/A	N/A		N/A	 	 		
Availability	Now	Now	Now	Now	Now	Now	Now	Now	Now	Now

 **Data** • Faster • Safer  Quantum Safe  FIPS 140 CMVP, NIST (Government)  SESIP/PSA  ISO 26262  Common Criteria

Source: company reports

However, we think Rambus faces intense competition in this attractive market, from two structurally advantaged groups.

- Leading AI ASIC design partners, such as Broadcom, Marvell, and MediaTek, possess extensive in-house expertise and proprietary portfolios of memory controllers and interconnect IP, leaving little room for third-party licensing within their hyperscaler programs.** These capabilities are the by-product of decades of shipping connectivity-intensive silicon at scale, hardened across thousands of tape-outs rather than developed for licensing. Broadcom refined its industry-leading SerDes and HBM PHY/ controller stack through successive generations of networking switch silicon and custom accelerator engagements, especially Google TPU program it has served across multiple generations. Marvell built its portfolio on storage and networking controller heritage and reinforced it through acquisition - Avera (the former IBM/ GlobalFoundries ASIC design unit) in 2019 and Inphi's high-speed interconnect franchise in 2021. MediaTek's high-volume smartphone SoC business has kept its DDR memory controllers silicon-proven at leading-edge nodes for over a decade. Qualcomm has internal heritage developed through smartphone SoC, but also acquired Alphawave in 2025, which provides high-speed connectivity IP to AI datacenter. Because memory controller and interconnect IP portfolios are the performance-critical portion of an AI ASIC, these vendors treat it as core differentiation for winning whole-chip engagements, resulting in bundling the IP within the design contract rather than sourcing it externally.
- Cadence and Synopsys benefit from strong ecosystem advantages, integrating their controllers, PHY, and interconnect IP offerings with industry-leading EDA tools and foundry design flows,** a bundling that standalone IP vendors cannot replicate. Rambus has already conceded ground where this bundling bites hardest, because it sold its SerDes and memory PHY IP business to Cadence for USD 110Mn in 2023 and exited the analog PHY layer to refocus on digital controller IP.

Given these dynamics, Rambus is primarily positioned to serve emerging or second-tier AI accelerator developers that lack internally developed IP and prefer to license complex memory controllers and interconnect technologies to accelerate product development. The customers Rambus showcases are precisely the emerging AI silicon designers we would expect: Etched, the transformer ASIC startup whose Sohu inference chip pairs a Rambus HBM memory controller with Rambus integration services to complete the memory subsystem; and MatX, the LLM accelerator startup founded by former Google engineers, which publicly endorsed the Mar 2026 HBM4E controller launch. Notably, no GPU or first-tier hyperscaler ASIC program appears among its disclosed IP engagements, with those sockets being locked up by the ASIC design partners' in-house IP. We believe this competitive backdrop explains why management is targeting only approximately 10% annual growth for the IP Licensing to

computing chips business, despite the rapid expansion of the broader AI ASIC industry.

Looking further ahead, we believe a more meaningful growth opportunity could emerge in edge AI over the mid to long term. Unlike the hyperscaler AI accelerator market, where a handful of CSPs concentrate design activities with selective leading ASIC partners, edge AI chip development is likely to be far more fragmented, spanning automotive, IoT, and industrial applications. A long tail of chip designers with limited in-house expertise creates structurally greater demand for outsourced memory controller and interconnect IP solutions, and Rambus's breadth across memory, interconnect, and security IP positions it well to capture this demand as the market matures.

EXHIBIT 11: Rambus has to compete with two groups of advantaged competitors in IP licensing for computing chips business, resulting in much slower segment growth expectation than the rapidly-expanding AI ASIC downstream market

WHAT RAMBUS LICENSES

● Memory controllers

HBM4 / HBM4E · GDDR7 · DDR5. Digital controllers for the AI memory subsystem — 100+ HBM controller design wins to date.

● Interconnect controllers

PCIe (to Gen 7) · CXL. High-speed data-path and memory-expansion IP.

● Security IP

Root of trust and crypto cores (CryptoManager; DPA countermeasures).

Built by acquisition · exited analog PHY

Cryptography Research '11 (security) · **AnalogX + PLDA '21** (PCIe/CXL) · **Hardent '22**.
Sold SerDes & memory-PHY IP to Cadence for **\$110M (2023)** — refocused on digital controllers, conceding the analog PHY layer.

TWO STRUCTURALLY ADVANTAGED GROUPS of COMPETITORS

AI ASIC design partners — in-house IP



Interface & memory IP hardened over decades and thousands of tape-outs, bundled inside whole-chip contracts — little room for third-party licensing in flagship hyperscaler programs

EDA incumbents — ecosystem bundling



Integrate interface & interconnect IP with leading EDA tools and foundry design flows — a bundling advantage standalone IP vendors cannot replicate

RAMBUS'S LANE — EMERGING & SECOND-TIER AI ACCELERATORS

Positioned for developers without in-house IP that license complex memory-controller / interconnect tech to speed time-to-market. No first-tier hyperscaler ASIC program appears among disclosed IP engagements.



Transformer-ASIC startup. Sohu Inference chip pairs a Rambus HBM controller with Rambus integration services.



LLM-accelerator startup (ex-Google). Publicly endorsed the Mar-2026 HBM4E Controller launch.

~10%

management's annual IP-licensing growth target — well below AI-ASIC market growth

Source: company reports, Bernstein analysis

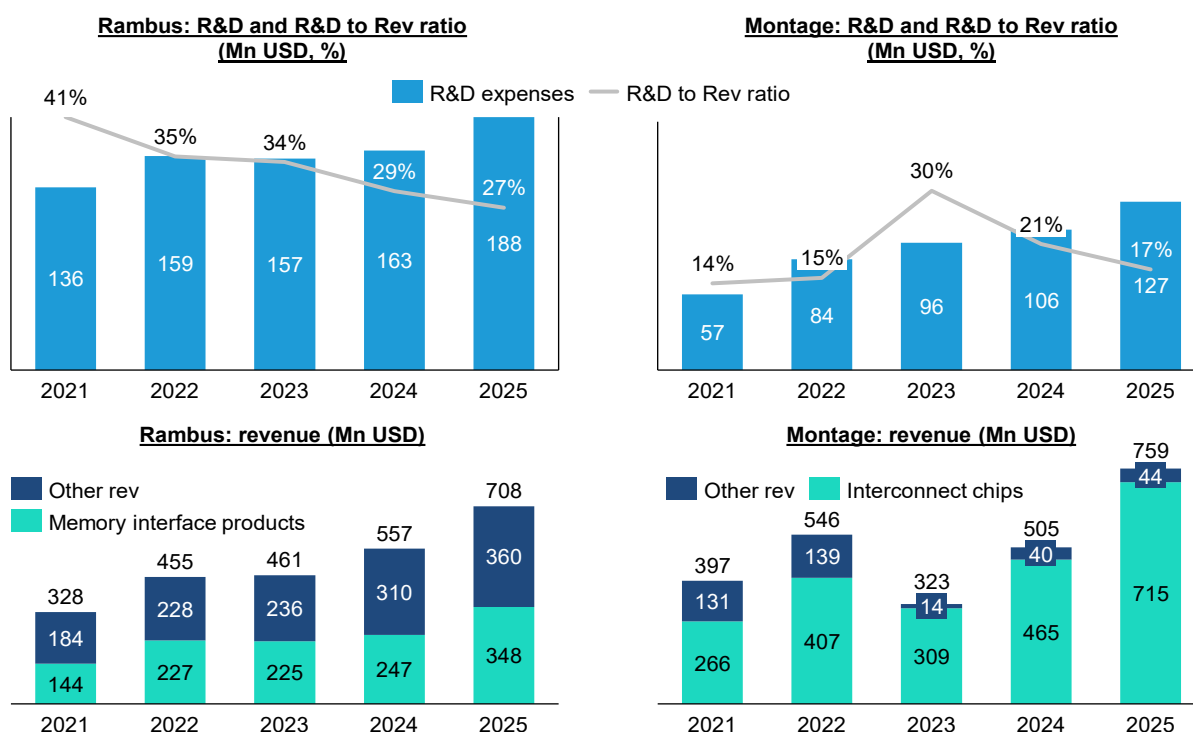
Nevertheless, royalties and IP licensing businesses in lower growth rate dilute the company's overall growth profile. The relatively low-growth licensing and contract businesses, two segments guided flat and ~10% annual growth respectively, still represent roughly half of total company revenue in 2025. As a result, we expect company-wide revenue growth to trail that of Montage, whose lower-growth businesses account for only an LSD percentage of total revenue.

MODERATE OPERATING LEVERAGE AND EARNINGS GROWTH RESTRICTED BY R&D INTENSITY

Rambus maintains a significantly broader technology portfolio than Montage, which results in annual R&D spending approximately 40-50% higher. Apart from the overlapping memory interface chips portfolio, the IP business alone spans memory controllers across DDR/LPDDR/GDDR/HBM, interconnect controllers across CXL and PCIe, video compression, and a full security stack from root of trust to quantum-safe cryptography. Each of these tech IP demands sustained R&D and engineering spending across data center, automotive, government, and IoT end markets. Rambus spent USD 188Mn for R&D in 2025, versus USD 127Mn for Montage.

Meanwhile, Rambus generates lower revenue relative to its larger R&D investment, leading to a substantially higher R&D-to-revenue ratio than Montage. We believe the lower revenue results from its lower market share in the key memory interface chip market, partly a consequence of the historical relationships with memory vendors discussed above. Exhibit 12 shows Rambus's R&D-to-revenue ratio stood at 27% in 2025, versus 17% for Montage. Given that R&D expenses account for more than 60% of Rambus's total operating expenses, R&D efficiency is the critical driver of long-term earnings growth. In our view, Rambus is likely to realize operating leverage more gradually than Montage as revenue scales.

EXHIBIT 12: **Rambus's broader tech portfolio and lower market share lead to higher R&D expenses and lower efficiency, compared to more focused Montage**



Source: Bloomberg, Bernstein analysis

DISCLOSURE APPENDIX

I. REQUIRED DISCLOSURES

References to "Bernstein" or the "Firm" in these disclosures relate to the following entities: Bernstein Institutional Services LLC (April 1, 2024 onwards), Sanford C. Bernstein & Co., LLC (pre April 1, 2024), Bernstein Autonomous LLP, BSG France S.A. (April 1, 2024 onwards), Sanford C. Bernstein (Hong Kong) Limited 盛博香港有限公司, Sanford C. Bernstein (Canada) Limited, Sanford C. Bernstein (India) Private Limited (SEBI registration no. INH000006378), Sanford C. Bernstein (Singapore) Private Limited, Sanford C. Bernstein Japan KK (サンフォード・C・バーンスタイン株式会社) and analysts employed by Société Générale Africa Technologies & Services to produce Bernstein research under a Global Services Agreement in place between Bernstein and Société Générale.

Bernstein is part of a joint venture between Société Générale (SG) and AllianceBernstein, L.P. (AB). Unless specifically noted otherwise, for purposes of these disclosures, references to Bernstein's "affiliates" relate to both SG and AB and their respective affiliates.

VALUATION METHODOLOGY**Montage Technology**

We set our **A-share TP at CNY 400**, based on **50x 2BF P/E** (27Q3–28Q2 earnings). We raise our 2027/2028 EPS estimates to CNY 5.68/10.82, respectively, on a stronger outlook for the CPU cycle and MRDIMM penetration. For the **H shares**, we set a target price of **HKD 520**, implying a **15% premium** to our A-share TP (we use the CNY to HKD exchange rate at 1:1.13). The H share premium reflects that global investors favor Montage as a scarce China AI-exposed name without direct geopolitical risks, unlike many other Chinese semiconductor companies that face entity list restrictions or export control headwinds. At our target price, the implied P/E multiple on 2BF EPS for H shares is 58x.

RISKS**Montage Technology**

Key potential risks:

1. A decrease in memory and AIDC server demand;
2. Intensify competition that leads to share loss and lower margin;
3. Failure to introduce new products for the AIDC networking market.

RATINGS DEFINITIONS, BENCHMARKS AND DISTRIBUTION**EQUITY RATINGS DEFINITIONS****Bernstein brand**

The Bernstein brand rates stocks based on forecasts of relative performance for the next 12 months versus the S&P 500 for stocks listed on the U.S. and Canadian exchanges, versus the Bloomberg Europe Developed Markets Large and Mid Cap Price Return Index EUR (EDME) for stocks listed on the European exchanges and emerging markets exchanges outside of the Asia Pacific region, versus the Bloomberg Japan Large and Mid Cap Price Return Index USD (JPL) for stocks listed on the Japanese exchanges, and versus the Bloomberg Asia ex-Japan Large and Mid Cap Price Return Index (ASIA) for stocks listed on the Asian (ex-Japan) exchanges -unless otherwise specified.

The Bernstein brand has three categories of ratings:

- Outperform: Stock will outpace the market index by more than 15 pp
- Market-Perform: Stock will perform in line with the market index to within +/- 15 pp
- Underperform: Stock will trail the performance of the market index by more than 15 pp

Coverage Suspended: Coverage of a company under the Bernstein research brand has been suspended. Ratings and price targets are suspended temporarily, are no longer current, and should therefore not be relied upon.

Not Rated: A rating assigned when the stock cannot be accurately valued, or the performance of the company accurately predicted, at the present time. The covering analyst may continue to publish research reports on the company to update investors on events and developments.

Not Covered (NC) denotes companies that are not under coverage.

Bernstein brand stock ratings are based on a 12-month time horizon.

Autonomous brand – common stocks

The Autonomous brand rates common stocks as indicated below. As our benchmarks we use the Bloomberg Europe 600 Financials Price Return Index (E600BK) and Bloomberg Europe Dev Mkt Financials Large and Mid Cap Price Ret Index EUR (EDMFI) index for developed European banks and Payments, the Bloomberg Europe 600 Insurance Price Return Index (E600IN) for European insurers, the S&P 500 and S&P Financials for US banks and Payments coverage, S5LIFE for US Insurance, the S&P Insurance Select Industry (SPSIINS) for US Non-Life Insurers coverage, the Bloomberg Emerging Markets Financials Large, Mid and Small Cap Price Return Index (EMLSF) for emerging market banks and insurers and Payments, and the Bloomberg Japan Financials Large & Mid Cap Price Return Index (JPFILM) for Japanese Financials. Ratings are stated relative to the sector (not the market).

The Autonomous brand has three categories of common stock ratings:

- **Outperform (OP):** Stock will outpace the relevant index by more than 10 pp
- **Neutral (N):** Stock will perform in line with the market index to within +/- 10 pp
- **Underperform (UP):** Stock will trail the performance of the relevant index by more than 10 pp

Coverage Suspended: Coverage of a company under the Autonomous research brand has been suspended. Ratings and price targets are suspended temporarily, are no longer current, and should therefore not be relied upon.

Not Rated: A rating assigned when the stock cannot be accurately valued, or the performance of the company accurately predicted, at the present time. The covering analyst may continue to publish research reports on the company to update investors on events and developments.

Those denoted as 'Feature' (e.g., Feature Outperform FOP, Feature Under Outperform FUP) are our core ideas.

Not Covered (NC) denotes companies that are not under coverage.

Autonomous brand common stock ratings are based on a 12-month time horizon.

Autonomous brand – preferred stocks

The Autonomous brand has three categories of preferred stock ratings:

- **Outperform (OP):** The total return of the preferred instrument is expected to outperform preferred securities of other issuers operating in similar sectors or rating categories over the next six months.
- **Neutral (N):** The total return of the preferred instrument is expected to perform in line with preferred securities of other issuers operating in similar sectors or rating categories over the next six months.
- **Underperform (UP):** The total return of the preferred instrument is expected to underperform preferred securities of other issuers operating in similar sectors or rating categories over the next six months.

Autonomous preferred stock ratings are based on a 6-month time horizon.

AUTONOMOUS CREDIT RESEARCH

Where this report contains investment recommendations for credit instruments, as defined in article 3(1)(35) of the Market Abuse Regulation, the information below is presented to comply with its disclosure requirements.

The report may also include reference(s) to published opinions by other Autonomous or Bernstein analysts covering the equity securities of the issuer(s) referenced herein. Please note an investment recommendation for credit instruments published by the author(s) of this report may differ from the published view of the analyst covering equity securities for the issuer(s) contained in this report and vice versa.

CREDIT RATINGS DEFINITIONS

The Autonomous brand has three categories of credit ratings:

- **Credit Outperform (C-OP):** The total return of the Reference Credit Instrument is expected to outperform the credit spread of bonds of other issuers operating in similar sectors or rating categories over the next six months.
- **Credit Neutral (C-N):** The total return of the Reference Credit Instrument is expected to perform in line with the credit spread of bonds of other issuers operating in similar sectors or rating categories over the next six months.
- **Credit Underperform (C-UP):** The total return of the Reference Credit Instrument is expected to underperform the credit spread of bonds of other issuers operating in similar sectors or rating categories over the next six months.

Autonomous credit ratings are based on a 6-month time horizon.

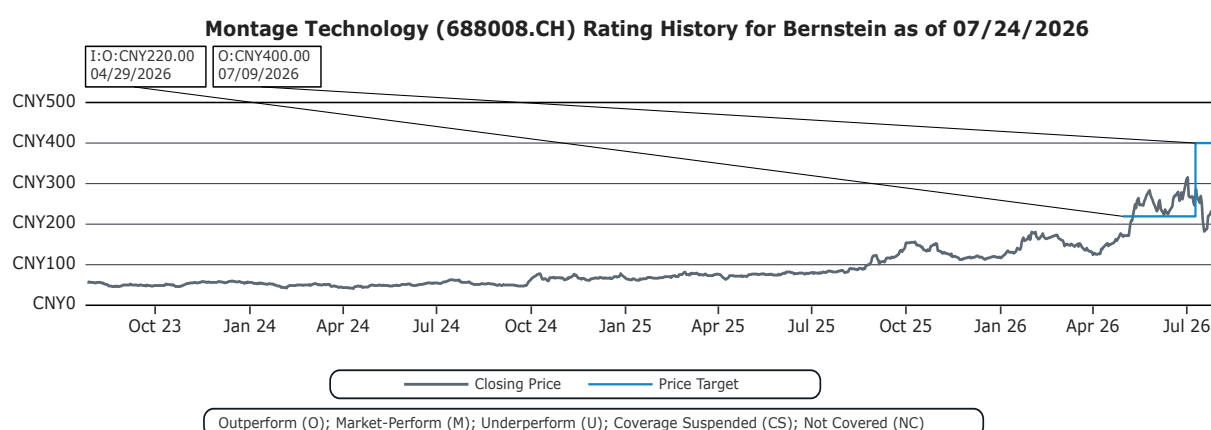
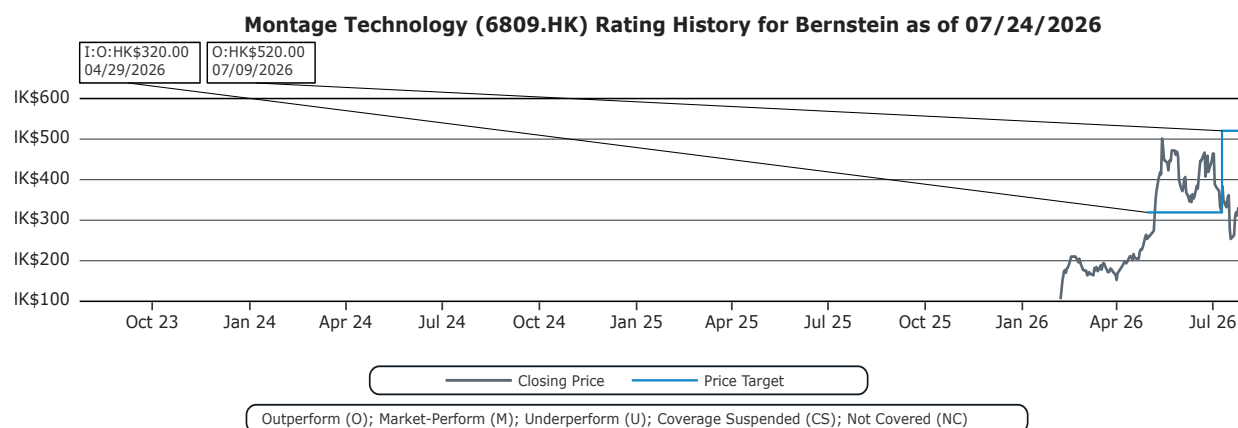
A list of all investment recommendations produced by the author(s) of this report alongside credit ratings history are available upon request.

It is at the sole discretion of the Firm as to when to initiate, update and cease research coverage. The Firm has established, maintains and relies on information barriers to control the flow of information contained in one or more areas (i.e. the private side) within the Firm, and into other areas, units, groups or affiliates (i.e. public side) of the Firm

DISTRIBUTION OF EQUITY RATINGS/INVESTMENT BANKING SERVICES

Equity Rating	Market Abuse Regulation (MAR) and FINRA Rating Category	Global Rating Distribution	Investment Banking Relationships*
Outperform	BUY	51.2%	15.3%
Market-Perform (Bernstein Brand) Neutral (Autonomous Brand)	HOLD	35.8%	16.2%
Underperform	SELL	13.1%	13.6%

* These figures represent the percentage of companies within each equity rating category for which affiliates of Bernstein have provided investment banking services within the previous 12 months.
As of June 30, 2026. All figures are updated quarterly.

PRICE CHARTS / RATINGS AND PRICE TARGET HISTORY

All price target and closing price data in the chart(s) above are denominated in the currency noted in the Ticker Table of this report.

OTHER MATTERS

The legal entity(ies) employing the analyst(s) listed in this report, and their location, can be determined by the country code of their phone number, as follows:

- +1 Bernstein Institutional Services LLC; New York, New York, USA
- +44 Bernstein Autonomous LLP; London UK
- +212 Société Générale Africa Technologies & Services; Casablanca, Morocco
- +33 BSG France S.A.; Paris, France
- +34 BSG France S.A.; Madrid, Spain
- +41 Bernstein Autonomous LLP; Geneva, Switzerland
- +49 BSG France S.A.; Frankfurt, Germany
- +91 Sanford C. Bernstein (India) Private Limited; Mumbai, India
- +852 Sanford C. Bernstein (Hong Kong) Limited 盛博香港有限公司; Hong Kong, China
- +65 Sanford C. Bernstein (Singapore) Private Limited; Singapore

+81 Sanford C. Bernstein Japan KK; Tokyo, Japan

Where this report has been prepared by research analyst(s) employed by a non-US affiliate, such analyst(s), is/are (unless otherwise expressly noted below) not registered as associated persons of Bernstein Institutional Services LLC or any other SEC-registered broker-dealer and are not licensed or qualified as research analysts with FINRA. Accordingly, such analyst(s) may not be subject to FINRA's restrictions regarding (among other things) communications by research analysts with a subject company, interactions between research analysts and investment banking personnel, participation by research analysts in solicitation and marketing activities relating to investment banking transactions, public appearances by research analysts, and trading securities held by a research analyst account.

Where this report has been prepared by research analyst(s) employed by Société Générale Africa Technologies & Services (part of the Société Générale group of companies), it has been prepared on behalf of a Bernstein company under a Global Services Agreement in place between Bernstein and Société Générale.

CERTIFICATION

Each research analyst listed in this report, who is primarily responsible for the preparation of the content of this report, certifies that all of the views expressed in this publication accurately reflect that analyst's personal views about any and all of the subject securities or issuers and that no part of that analyst's compensation was, is, or will be, directly or indirectly, related to the specific recommendations or views in this publication.

II. ADDITIONAL GLOBAL CONFLICT DISCLOSURES

It is at the sole discretion of the Firm as to when to initiate, update and cease research coverage. The Firm has established, maintains and relies on information barriers to control the flow of information contained in one or more areas (i.e., the private side) within the Firm, and into other areas, units, groups or affiliates (i.e., public side) of the Firm.

III. OTHER IMPORTANT INFORMATION AND DISCLOSURES

Separate branding is maintained for "Bernstein" and "Autonomous" research products.

- Bernstein produces a number of different types of research products including, among others, fundamental analysis and quantitative analysis under both the "Autonomous" and "Bernstein" brands. Recommendations contained within one type of research product may differ from recommendations contained within other types of research products, whether as a result of differing time horizons, methodologies or otherwise. Furthermore, views or recommendations within a research product issued under one brand may differ from views or recommendations under the same type of research product issued under the other brand. The Research Ratings System for the two brands and other information related to those Rating Systems are included in the previous section.
- Autonomous operates as a separate business unit within the following entities: Bernstein Institutional Services LLC, Bernstein Autonomous LLP, Sanford C. Bernstein (Hong Kong) Limited 盛博香港有限公司 and Sanford C. Bernstein (India) Private Limited. For information relating to "Autonomous" branded products (including certain Sales materials) please visit: www.autonomous.com. For information relating to Bernstein branded products please visit: www.bernsteinresearch.com.

Analysts are compensated based on aggregate contributions to the research franchise as measured by account penetration, productivity and proactivity of investment ideas. No analysts are compensated based on performance in, or contributions to, generating investment banking revenues.

This report has been produced by an independent analyst as defined in Article 3 (1)(34)(i) of EU 596/2014 Market Abuse Regulation ("MAR") and the same article of MAR as it forms part of United Kingdom domestic law by virtue of the European Union (Withdrawal) Act 2018.

To our readers in the United States: Bernstein Institutional Services LLC, a broker-dealer registered with the U.S. Securities and Exchange Commission ("SEC") and a member of the U.S. Financial Industry Regulatory Authority, Inc. ("FINRA") is distributing this publication in the United States and accepts responsibility for its contents. Where this material contains an analysis of debt product(s), such material is intended only for institutional investors and is not subject to the US independence and disclosure standards applicable to debt research prepared for retail investors.

Bernstein Institutional Services LLC may act as principal for its own account or as agent for another person (including an affiliate) in sales or purchases of any security which is a subject of this report. This report does not purport to meet the objectives or needs of any specific individuals, entities or accounts.

To our readers in Canada: If this publication pertains to a Canadian domiciled company, it is being distributed in Canada by Sanford C. Bernstein (Canada) Limited, which is licensed and regulated by the Canadian Investment Regulatory Organization. If the publication pertains to a non-Canadian domiciled company, it is being distributed by Bernstein Institutional Services LLC, which is licensed and regulated by both the SEC and FINRA, into Canada under the International Dealers Exemption.

This document may not be passed onto any person in Canada unless that person qualifies as "permitted client" as defined in Section 1.1 of NI 31-103.

To our readers in Brazil: This report has been prepared by Bernstein Institutional Services LLC, and Banco BTG Pactual S.A. ("BTG") is responsible for the distribution of this report in Brazil.

To readers in the United Kingdom: This publication has been issued or approved for issue in the United Kingdom by Bernstein Autonomous LLP, authorised and regulated by the Financial Conduct Authority and located at 60 London Wall, London EC2M 5SH, +44 (0)20-7170-5000. Registered in England & Wales No OC343985.

This document is for distribution only to persons who (i) have professional experience in matters relating to investments falling within Article 19(5) of the Financial Services and Markets Act 2000 (Financial Promotion) Order 2005 (as amended, the "Financial Promotion Order"), (ii) are persons falling within Article 49(2)(a) to (d) ("high net worth companies, unincorporated associations, etc.") of the Financial Promotion Order, (iii) are outside the United Kingdom, or (iv) are persons to whom an invitation or inducement to engage in investment activity (within the meaning of section 21 of the FSMA) in connection with the issue or sale of any securities may otherwise lawfully be communicated or caused to be communicated (all such persons together being referred to as "relevant persons"). This document is directed only at relevant persons and must not be acted on or relied on by persons who are not relevant persons. Any investment or investment activity to which this document relates is available only to relevant persons and will be engaged in only with relevant persons.

To our readers in the member states of the EEA: This publication is being distributed by BSG France SA, which is authorised and regulated by the Autorité de Contrôle Prudentiel et de Résolution (ACPR) and Autorité des Marchés Financiers (AMF).

To our readers in Hong Kong: This publication is being distributed in Hong Kong by Sanford C. Bernstein (Hong Kong) Limited 盛博香港有限公司, which is licensed and regulated by the Hong Kong Securities and Futures Commission (Central Entity No. AXC846) to carry out Type 4 (Advising on Securities) regulated activities and subject to the licensing conditions mentioned in the SFC Public Register (<https://www.sfc.hk/publicregWeb/corp/AXC846/details>). This publication is solely for professional investors, as defined in the Securities and Futures Ordinance (Cap. 571). The purpose of this report is solely to provide an analysis of the issuers referred to in this report and is not intended for any purpose contrary to the laws of Hong Kong.

To our readers in Singapore: This publication is being distributed in Singapore by Sanford C. Bernstein (Singapore) Private Limited, only to accredited investors or institutional investors, as defined in the Securities and Futures Act 2001 of Singapore ("SFA"). Recipients in Singapore should contact Sanford C. Bernstein (Singapore) Private Limited in respect of matters arising from, or in connection with, this publication. Sanford C. Bernstein (Singapore) Private Limited is regulated by the Monetary Authority of Singapore and licensed under the SFA as a capital markets services licence holder for dealing in capital markets products that are securities and collective investment schemes and an exempt financial adviser for advising on, issuing and promulgating analyses and reports on securities. Sanford C. Bernstein (Singapore) Private Limited is registered in Singapore with Company Registration No. 20213710W and located at 8 Marina Boulevard, #12-01, Marina Bay Financial Centre, Singapore 018981, +65-6326-7000.

To our readers in the People's Republic of China: The securities referred to in this document are not being offered or sold and may not be offered or sold, directly or indirectly, in the People's Republic of China (for such purposes, not including the Hong Kong and Macau Special Administrative Regions or Taiwan, the "PRC") in contravention of any applicable laws of the PRC.

This document does not constitute an offer to sell or the solicitation of an offer to buy any securities in the PRC to any person to whom it is unlawful to make the offer or solicitation in the PRC.

We do not represent that this document may be lawfully distributed, or that any securities may be lawfully offered, in compliance with any applicable registration or other requirements in the PRC, or pursuant to an exemption available thereunder, or assume any responsibility for facilitating any such distribution or offering. In particular, no action has been taken by us which would permit a public offering of any securities or distribution of this document in the PRC. Accordingly, the securities are not being offered or sold within the PRC by means of this document or any other document. Neither this document nor any advertisement or other offering material may be distributed or published in the PRC, except under circumstances that will result in compliance with any applicable laws and regulations.

To our readers in Japan: This publication is being distributed in Japan by Sanford C. Bernstein Japan KK (サンフォード・C・バーンスタイン株式会社), which is registered in Japan as a Financial Instruments Business Operator with the Kanto Local Finance Bureau (registration number: The Director-General of Kanto Local Finance Bureau (FIBO) No.3387) and regulated by the Financial Services Agency. It is also a member of Investment Management Association of Japan. This publication is solely

for qualified institutional investors in Japan only, as defined in Article 2, paragraph (3), items (i) of the Financial Instruments and Exchange Act.

For the institutional client readers in Japan who have been granted access to the Bernstein website by Daiwa Securities Group Inc. ("Daiwa"), your access to this document should not be construed as meaning that Bernstein is providing you with investment advice for any purposes. Whilst Bernstein has prepared this document, your relationship is, and will remain with, Daiwa, and Bernstein has neither any contractual relationship with you nor any obligations towards you.

To our readers in Australia: Sanford C. Bernstein (Hong Kong) Limited 盛博香港有限公司 is responsible for distributing research in Australia. It is regulated by the Securities and Exchange Commission under U.S. laws, by the Financial Conduct Authority under U.K. laws, which differs from Australian laws. Sanford C. Bernstein (Hong Kong) Limited 盛博香港有限公司 is exempt from the requirement to hold an Australian financial services license under the Corporations Act 2001 in respect of the provision of the following financial services to wholesale clients:

- providing financial product advice;
- dealing in a financial product;
- making a market for a financial product; and
- providing a custodial or depository service.

To our readers in India: This publication is being distributed in India by Sanford C. Bernstein (India) Private Limited (SCB India) which is licensed and regulated by Securities and Exchange Board of India ("SEBI") as a research analyst entity under the SEBI (Research Analyst) Regulations, 2014, having registration no. INH000006378 and as a stock broker having registration no. INZ000213537. SCB India is currently engaged in the business of providing research and stock broking services. Please refer to www.bernsteinresearch.in for more information.

- SCB India is a Private limited company incorporated under the Companies Act, 2013, on April 12, 2017 bearing corporate identification number U65999MH2017FTC293762, and registered office at Level 3A, 4th Floor, First International Financial Centre, Plot Nos C-54 and C-55, G Block, Near CBI Office, Bandra Kurla Complex, Bandra (East), Mumbai 400098, Maharashtra, India (Phone No: +91-22-68421401).
- For details of Associates (i.e., affiliates/group companies) of SCB India, kindly email MUM-BERNSTEIN-InCompliance@bernsteinsg.com.
- SCB India does not have any disciplinary history as on the date of this report.
- Except as noted above, SCB India and/or its Associates (i.e., affiliates/group companies), the Research Analysts authoring this report, and their relatives
 - do not have any financial interest in the subject company
 - do not have actual/beneficial ownership of one percent or more in securities of the subject company;
 - is not engaged in any investment banking activities for Indian companies, as such;
 - have not managed or co-managed a public offering in the past twelve months for any Indian companies;
 - have not received any compensation for investment banking services or merchant banking services from the subject company in the past 12 months;
 - have not received compensation for brokerage services from the subject company in the past twelve months;
 - have not received any compensation or other benefits from the subject company or third party related to the specific recommendations or views in this report; and
 - do not currently, but may in the future, act as a market maker in the financial instruments of the companies covered in the report.

- do not have any conflict of interest in the subject company as of the date of this report.
- Except as noted above, the subject company has not been a client of SCB India during twelve months preceding the date of distribution of this research report. Neither SCB India nor its Associates (i.e., affiliates/group companies) have received compensation for products or services other than investment banking, merchant banking or brokerage services from the subject company in the past twelve months.
- The principal research analyst(s) who prepared this report, members of the analysts' team, and members of their households are not an officer, director, employee or advisory board member of the companies covered in the report.
- Our Compliance officer / Grievance officer is Ms. Rupal Talati, who can be reached at +91-22-68421451, or MUM-BERNSTEIN-InCompliance@bernsteinsg.com / Scbin-investorgrievance@bernsteinsg.com
- The Research investor charter and Terms & Conditions of SCB India are available on its website and may be accessed at [Sanford C. Bernstein \(India\) Private Limited](https://bernsteinresearch.in/) (<https://bernsteinresearch.in/>) for your reference.
- Disclaimer: Registration granted by SEBI, and certification from NISM, is in no way a guarantee of performance of the intermediary or provide any assurance of returns to investors. Investments in securities market are subject to market risks. Read all the related documents carefully before investing.

To our readers in Switzerland: This document is provided in Switzerland by or through Bernstein Autonomous LLP, and is provided only to qualified investors as defined in article 10 of the Swiss Collective Investment Scheme Act ("CISA") and related provisions of the Collective Investment Scheme Ordinance and in strict compliance with applicable Swiss law and regulations. The products mentioned in this document may not be suitable for all types of investors. This document is based on the Directives on the Independence of Financial Research issued by the Swiss Bankers Association (SBA) in January 2008.

To our readers in the Middle East: Bernstein Autonomous LLP, DIFC branch has its principal office at Gate Village 06, DIFC, Dubai, UAE. Bernstein Autonomous LLP, DIFC branch is regulated by the Dubai Financial Services Authority (DFSA) with the registration number CL10040 and is provisioned for Arranging Deals in Investments and Advising on Financial Products. All communications and services are directed at Professional Clients and Market Counterparties only (as defined in the DFSA rulebook). Persons other than Professional Clients and Market Counterparties, such as Retail Clients, are not the intended recipients of our communications or services.

LEGAL

All research publications are disseminated to our clients through posting on the firm's password protected websites, bernsteinresearch.com and autonomous.com. Certain, but not all, research publications are also made available to clients through third-party vendors or redistributed to clients through alternate electronic means as a convenience.

This publication has been published and distributed in accordance with the Firm's policy for management of conflicts of interest in investment research, a copy of which is available from Bernstein Institutional Services LLC, Director of Compliance, 245 Park Avenue, New York, NY 10167. Additional disclosures and information regarding Bernstein's business are available on our website www.bernsteinresearch.com.

The securities described herein may not be eligible for sale in all jurisdictions or to certain categories of investors where that permission profile is not consistent with the licenses held by the entities noted herein. This document is for distribution only as may be permitted by law. This publication is not directed to, or intended for distribution to or use by, any person or entity who is a citizen or resident of, or located in any locality, state, country or other jurisdiction where such distribution, publication, availability or use would be contrary to law or regulation or which would subject any of the entities referenced herein or any of their subsidiaries or affiliates to any registration or licensing requirement within such jurisdiction. This publication is based upon public sources we believe to be reliable, but no representation is made by us that the publication is accurate or complete. We do not undertake to advise you of any change in the reported information or in the opinions herein. This publication was prepared and issued by entity referred to herein for distribution to eligible counterparties or professional clients. This publication is not an offer to buy or sell any security, and it does not constitute investment, legal or tax advice. The investments referred to herein may not be suitable for you. Investors must make their own investment decisions in consultation with their professional advisors in light of their specific circumstances. The value of investments may fluctuate, and investments that are denominated in foreign currencies may fluctuate in value as a result of exposure to exchange rate movements. Information about past performance of an investment is not necessarily a guide to, indicator of, or assurance of, future performance.

This report is directed to and intended only for our clients who are "eligible counterparties", "professional clients", "institutional investors" and/or "professional investors" as defined by the aforementioned regulators, and must not be redistributed to retail

clients as defined by the aforementioned regulators. Retail clients who receive this report should note that the services of the entities noted herein are not available to them and should not rely on the material herein to make an investment decision. The result of such act will not hold the entities noted herein liable for any loss thus incurred as the entities noted herein are not registered/ authorised/ licensed to deal with retail clients and will not enter into any contractual agreement/arrangement with retail clients. This report is provided subject to the terms and conditions of any agreement that the clients may have entered into with the entities noted herein. All research reports are disseminated on a simultaneous basis to eligible clients through electronic publication to our client portal.

The information in this report was prepared by Bernstein solely for the internal business use of our clients. Clients may store, display, analyze, reformat and print the information in this report for this limited use only. Clients may not copy, alter, create derivative works, resell, reverse engineer, commercially exploit, share or distribute any part of the information contained herein for any purpose without Bernstein's express written consent. These restrictions include extracting data or using the content to develop indices or other products. Further, you may not use this report, or any portion of this report, to train or finetune any third-party machine learning or artificial intelligence system, or as a prompt or input into any such system. You also may not, without Bernstein's express written consent, do any of the foregoing in connection with your own internal machine learning or artificial intelligence system.

Bernstein may use artificial intelligence tools in the preparation of its materials. Any such materials are reviewed by Bernstein's research analysts prior to publication.

This report has been prepared for information purposes only and is based on current public information that we consider reliable, but the entities noted herein do not warrant or represent (express or implied) as to the sources of information or data contained herein are accurate, complete, not misleading or as to its fitness for the purpose intended even though the entities noted herein rely on reputable or trustworthy data providers, it should not be relied upon as such. Opinions expressed are the author(s)' current opinions as of the date appearing on the material only and we do not undertake to advise you of any change in the reported information or in the opinions herein.

Any references to Societe Generale herein are purely factual, based upon publicly available information, and included for comparative purposes only. They do not constitute an opinion or recommendation with respect to the securities of Societe Generale.

This publication was prepared and issued by the entity referred to herein for distribution to eligible counterparties or professional clients. The information in this report is intended for general circulation and does not constitute an offer to buy or sell any security, investment, legal or tax advice nor a personal recommendation, as defined by any of the aforementioned regulators. It does not take into account the particular investment objectives, financial situations, or needs of individual investors. The report has not been reviewed by any of the aforementioned regulators and does not represent any official recommendation from the aforementioned regulators. The investments referred to herein may not be suitable for you. Investors must make their own investment decisions in consultation with advice sought from a financial adviser regarding the suitability of the investment product, taking into account the specific investment objectives, financial situation or particular needs of any recipient of the recommendation, before the recipient makes a commitment to purchase the investment product.

The analysis contained herein is based on numerous assumptions. Different assumptions could result in materially different results. The information in this report does not constitute, or form part of, any offer to sell or issue, or any offer to purchase or subscribe for shares, or to induce engagement in any other investment activity. The value of any securities or financial instruments mentioned in this report may fluctuate subject to market conditions. Information about past performance of an investment is not necessarily a guide to, indicative of, or assurance of future performance. Estimates of future performance mentioned by the research analyst in this report are based on assumptions that may not be realized due to unforeseen factors like market volatility/fluctuation. In relation to securities or financial instruments denominated in a foreign currency other than the clients' home currency, movements in exchange rates will have an effect on the value, either favorable or unfavorable. Before acting on any recommendations in this report, recipients should consider the appropriateness of investing in the subject securities or financial instruments mentioned in this report and, if necessary, seek independent professional advice.

The securities described herein may not be eligible for sale in all jurisdictions or to certain categories of investors where that permission profile is not consistent with the licenses held by the entities noted herein. This document is for distribution only as may be permitted by law. It is not directed to, or intended for distribution to or use by, any person or entity who is a citizen or resident of or located in any locality, state, country or other jurisdiction where such distribution, publication, availability or use would be contrary to law or regulation or would subject the entities noted herein to any regulation or licensing requirement within such jurisdiction.

Source: Bloomberg Index Services Limited. BLOOMBERG® is a trademark and service mark of Bloomberg Finance L.P. and its affiliates (collectively "Bloomberg"). Bloomberg or Bloomberg's licensors own all proprietary rights in the Bloomberg Indices. Neither Bloomberg nor Bloomberg's licensors approves or endorses this material, or guarantees the accuracy or completeness of any information herein, or makes any warranty, express or implied, as to the results to be obtained therefrom and, to the maximum

extent allowed by law, neither shall have any liability or responsibility for injury or damages arising in connection therewith.

No part of this material may be reproduced, distributed or transmitted or otherwise made available without prior consent of the entities noted herein. Copyright Bernstein Institutional Services LLC Bernstein Autonomous LLP, BSG France S.A., Sanford C. Bernstein (Hong Kong) Limited 盛博香港有限公司, Sanford C. Bernstein (Canada) Limited, Sanford C. Bernstein (India) Private Limited (SEBI registration no. INH000006378), Sanford C. Bernstein (Singapore) Private Limited and Sanford C. Bernstein Japan KK (サンフォード・C・バーンスタイン株式会社). All rights reserved. The trademarks and service marks contained herein are the property of their respective owners. Any unauthorized use or disclosure is strictly prohibited. The entities noted herein may pursue legal action if the unauthorized use results in any defamation and/or reputational risk to the entities noted herein and research published under the Bernstein and Autonomous brands.